13 次单位根的根式解

参数与标准素分解

$p = 13$，$h = (p-1)/2 = 6$，模 13 的最小原根 $g = 2$。

h 的标准素分解：$h = 3 \times 2$，子群链共 2 层。

$m_0 = 1$，$m_1 = 3$，$m_2 = 6$。

各层子群 $H_j = \{\pm g^{m_j t}\}$，其中 H_0 是全部 12 个非零剩余：

$$H_1 = \{1, 5, 8, 12\}$$

$$H_2 = \{1, 12\}$$

各层周期 $\eta_a^{(j)} = \sum_{b \in H_j} \zeta^{ab}$（$a$ 取陪集代表）。

各层周期与分解表

第 0 层：1 个周期

H_0 是全部 12 个非零剩余，故 $\eta_1^{(0)} = -1$。

第 1 层：3 个周期

$$\eta_1^{(1)} = \zeta^1 + \zeta^5 + \zeta^8 + \zeta^{12}$$

$$\eta_2^{(1)} = \zeta^2 + \zeta^3 + \zeta^{10} + \zeta^{11}$$

$$\eta_4^{(1)} = \zeta^4 + \zeta^6 + \zeta^7 + \zeta^9$$

第 2 层：6 个周期

$$\eta_1^{(2)} = \zeta^1 + \zeta^{12}$$

$$\eta_2^{(2)} = \zeta^2 + \zeta^{11}$$

$$\eta_3^{(2)} = \zeta^3 + \zeta^{10}$$

$$\eta_4^{(2)} = \zeta^4 + \zeta^9$$

$$\eta_5^{(2)} = \zeta^5 + \zeta^8$$

$$\eta_6^{(2)} = \zeta^6 + \zeta^7$$

分解表：每个母周期拆成 ℓ_j 个子周期

$$\eta_1^{(0)} = \eta_1^{(1)} + \eta_2^{(1)} + \eta_4^{(1)}$$

（第 1 层，$\ell = 3$。）

$$\eta_1^{(1)} = \eta_1^{(2)} + \eta_5^{(2)}$$

$$\eta_2^{(1)} = \eta_2^{(2)} + \eta_3^{(2)}$$

$$\eta_4^{(1)} = \eta_4^{(2)} + \eta_6^{(2)}$$

（第 2 层，$\ell = 2$。）

辅助单位根

$\ell = 2$ 时 $\omega_2^1 = -1$。
$\ell \geq 3$ 时按主支：

$$\omega_\ell^k = \frac{\Lambda_{\bar{k}}^{(\ell)}}{2} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{2\bar{k}}^{(\ell)}}\, i$$

其中 $\varepsilon_k = \operatorname{sgn}\sin(2\pi k/\ell)$，$\bar{k} = \min(k, \ell - k)$。

辅助问题 $p = 3$（供 ω_3^k 使用）

$h = 1$，$g = 2$，层阶 1。

$$\Lambda_1^{(3)} = -1$$

$\omega_3^k = \frac{1}{2}\Lambda_{\bar{k}}^{(3)} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{2\bar{k}}^{(3)}}\, i$。

逐层开方

每一步把母周期 $\eta_a^{(j-1)}$ 分解成 ℓ 个子周期 $\beta_t^{(a)}$。
预解式取离散 Fourier 变换：

$$\theta_{a,k}^{(j)} = \sum_{t=0}^{\ell-1} \omega_\ell^{-kt}\, \beta_t^{(a)}$$

$$\left(\theta_{a,k}^{(j)}\right)^\ell = u_{a,k}^{(j)} = \tau^k\left(u_{a,1}^{(j)}\right)$$

$$\eta_{b(t)}^{(j)} = \frac{1}{\ell}\left(\eta_a^{(j-1)} + \sum_{k=1}^{\ell-1} \omega_\ell^{kt}\, \theta_{a,k}^{(j)}\right)$$

（τ 把 ω_ℓ 送到 ω_ℓ^k；$b(t)$ 为第 t 个子周期。）
开方一律取正实主支；预解式的分支由因子 ω_ℓ^s 记录（见报告里的分支表）。

第 1 步（$\ell = 3$，**1 个母周期**）

$$u_{1,1}^{(1)} = \frac{-65 + 39i\sqrt{3}}{2}$$

$$\theta_{1,1}^{(1)} = \frac{\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}{4}$$

第 2 步（$\ell = 2$，**3 个母周期**）

$$u_{1,1}^{(2)} = \frac{104 - 8\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 8\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 3\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 3\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 6\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 6\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}{24}$$

$$\theta_{1,1}^{(2)} = \frac{\sqrt{2496 - 192\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 192\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 144\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 144\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{24}$$

$$u_{2,1}^{(2)} = \frac{104 - 12\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 12\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 4\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 4\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 3\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 3\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}{24}$$

$$\theta_{2,1}^{(2)} = \frac{\sqrt{2496 - 288\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 288\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{24}$$

$$u_{4,1}^{(2)} = \frac{52 - 3\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 3\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 3\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 3\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 2\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 2\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}{12}$$

$$\theta_{4,1}^{(2)} = \frac{\sqrt{624 - 36\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 24\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 24\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{12}$$

实部数值

$$\Lambda_1 = 2\cos\frac{2\pi\cdot 1}{13} = \frac{-8 + 2\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + 2\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \sqrt{2496 - 192\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 192\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 144\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 144\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

$$\Lambda_2 = 2\cos\frac{2\pi\cdot 2}{13} = \frac{-8 + \left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \sqrt{2496 - 288\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 288\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

$$\Lambda_3 = 2\cos\frac{2\pi\cdot 3}{13} = \frac{-8 + \left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - \sqrt{2496 - 288\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 288\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

$$\Lambda_4 = 2\cos\frac{2\pi\cdot 4}{13} = \frac{-8 + \left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} + 2\sqrt{624 - 36\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 24\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 24\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

$$\Lambda_5 = 2\cos\frac{2\pi\cdot 5}{13} = \frac{-8 + 2\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + 2\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - \sqrt{2496 - 192\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 192\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 144\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 144\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

$$\Lambda_6 = 2\cos\frac{2\pi\cdot 6}{13} = \frac{-8 + \left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 2\sqrt{624 - 36\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 36\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 24\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 24\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}$$

全部单位根

最终形式：

$$\omega_n^t = \frac{\Lambda_{\bar{t}}}{2} + \frac{\varepsilon_t}{2}\sqrt{2 - \Lambda_{\overline{2t}}}\,i, \qquad \Lambda_{\bar{t}} = 2\cos\frac{2\pi\bar{t}}{n}$$

其中 $\bar{t} = \min(t, n-t)$，$\varepsilon_t = \operatorname{sgn}\sin(2\pi t/n)$。

以下逐一写开（全部为具体根式）：

$$\omega_{n}^{t} = \frac{-8 + 2\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + 2\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \sqrt{2496 - 192\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 192\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 144\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 144\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48} + \frac{\varepsilon_t}{2}\sqrt{2 - \frac{-8 + \left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \sqrt{2496 - 288\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 288\left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 96\left(1 - 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 72\left(1 + 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 72\left(1 - 2i\sqrt{3} + 3ii\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{48}}\,i$$

